

YEAR 7 • UNIT 1 • EXTRA HOMEWORK

The Two That Do the Work

Multiplication and division • the engine room of Unit 1

Name: _____

Class: _____

Date given: _____

Date due: _____

Why this sheet exists

Everything in Unit 1 sits on top of two operations. A **factor** is found by **dividing**, factor **pairs** by **multiplying**, and the **HCF** simplifies a fraction by **dividing** top and bottom. So the faster and more surely you can multiply and divide, the easier the whole unit gets. That is all this sheet is: **extra practice on the two operations that do the work.**

Do it without a calculator, and show your working — working you can see is working you can check.

Part 1 • The facts worth knowing by heart

E1 Fill in the missing number

These are the ones that come up again and again in factors and divisibility. Aim to know them without counting.

(a) $6 \times 7 =$ _____

(f) $8 \times 8 =$ _____

(k) $8 \times 12 =$ _____

(b) $7 \times 8 =$ _____

(g) $12 \times 7 =$ _____

(l) $9 \times 12 =$ _____

(c) $8 \times 9 =$ _____

(h) $11 \times 9 =$ _____

(m) $7 \times 9 =$ _____

(d) $9 \times 6 =$ _____

(i) $9 \times 9 =$ _____

(n) $11 \times 11 =$ _____

(e) $7 \times 7 =$ _____

(j) $12 \times 12 =$ _____

(o) $7 \times 11 =$ _____

E2 Multiply

Set each one out properly and show your working in the box.

(a) Two digits by one digit.

34×6	47×8	58×7	93×4

(b) Three digits by one digit.

236×7	418×6	364×8

(c) Two digits by two digits.

23×14	36×25	48×17

Part 2 · Division

E3 Divide – these all come out exactly

A check that costs five seconds

When you finish a division, **multiply your answer back**. If $72 \div 6 = 12$, then 12×6 must come to 72. If it does not, you have just caught your own mistake.

(a) $72 \div 6 =$ _____

(e) $156 \div 12 =$ _____

(b) $96 \div 4 =$ _____

(f) $245 \div 5 =$ _____

(c) $91 \div 7 =$ _____

(g) $372 \div 4 =$ _____

(d) $126 \div 9 =$ _____

(h) $651 \div 3 =$ _____

E4 Divide – these do not

Word help – the words that describe what is left

exactly nothing is left over. $72 \div 6 = 12$ exactly.

remainder what is left over when it does not come out exactly. We write $53 \div 6 = 8$ r 5.

goes into “6 goes into 24 four times” means $24 \div 6 = 4$.

left over the everyday English for a remainder.

Write each answer as a whole number and a remainder, like this: $53 \div 6 = 8 \text{ r } 5$.

(a) $74 \div 8 =$ _____

(d) $137 \div 5 =$ _____

(b) $100 \div 7 =$ _____

(e) $191 \div 6 =$ _____

(c) $85 \div 9 =$ _____

(f) $250 \div 8 =$ _____

Part 3 • Putting them together

E5 Find the missing number

Multiplying and dividing undo each other

If $7 \times 9 = 63$, then $63 \div 7 = 9$ **and** $63 \div 9 = 7$. Every multiplication fact is also **two** division facts — which is why Part 1 is worth knowing by heart.

(a) $7 \times$ _____ $= 63$

(e) $6 \times$ _____ $= 84$

(b) _____ $\times 8 = 96$

(f) _____ $\times 9 = 117$

(c) $54 \div$ _____ $= 6$

(g) $132 \div$ _____ $= 11$

(d) _____ $\div 7 = 8$

(h) _____ $\div 4 = 25$

E6 Use it: is it a factor?

Straight out of Homework 3

A number is a **factor** of another if it divides into it **exactly**. So the way to find out is to **do the division and look at the remainder**.

Do each division, then circle **yes** or **no**. **Two of these you have already worked out on this sheet** — look back rather than starting again.

Question	The division	Is it a factor?
Is 6 a factor of 84?		yes / no
Is 8 a factor of 92?		yes / no
Is 9 a factor of 126?		yes / no
Is 7 a factor of 100?		yes / no

E7 Word problems**Word help – which operation does the sentence want?**

each / every	usually multiply. 6 boxes with 24 in each is 6×24 .
share equally between	divide.
how many are left	the remainder.
altogether / in total	the whole amount, once you have finished.

Write the calculation first, then the answer.

- 1. The markers.** Mr Bowen buys **6 boxes** of markers. There are **24 markers** in each box. How many markers does he have altogether?

- 2. The paper.** Mr Bowen shares **96 sheets of paper** equally between **8 students**. How many sheets does each student get?

- 3. The pizzas.** Mr Bowen orders **8 identical pizzas** for the class. Each pizza is cut into **12 slices**. There are **29 students**. Every student gets the same whole number of slices, and Mr Bowen eats the rest. How many slices does each student get, and how many does Mr Bowen eat?

Answer Key – Teacher Copy

E1 Fill in the missing number

- (a) 42 (b) 56 (c) 72 (d) 54 (e) 49 (f) 64 (g) 84 (h) 99
 (i) 81 (j) 144 (k) 96 (l) 108 (m) 63 (n) 121 (o) 77

Time this one rather than marking it: if a student can do all fifteen in under two minutes, the tables are not what is slowing them down in Unit 1 and Part 2 is where to look instead. The three worth re-drilling if they go wrong are (b) 7×8 , (c) 8×9 and (m) 7×9 – those three account for most of the errors in factor lists.

E2 Multiply

- (a) $34 \times 6 = 204$ $47 \times 8 = 376$ $58 \times 7 = 406$ $93 \times 4 = 372$
 (b) $236 \times 7 = 1652$ $418 \times 6 = 2508$ $364 \times 8 = 2912$
 (c) $23 \times 14 = 322$ $36 \times 25 = 900$ $48 \times 17 = 816$

The error to look for is carrying, not the tables: in (b), 418×6 needs a carry at every column. In (c), check they have written *two* rows of working and added them – a single row means they have multiplied by the units digit only, and that shows up as an answer roughly a tenth of the right size.

E3 Divide – these all come out exactly

- (a) 12 (b) 24 (c) 13 (d) 14 (e) 13 (f) 49 (g) 93 (h) 217

(c) $91 \div 7$ is the one that surprises people – 91 looks prime and is not. (h) $651 \div 3$ is the longest and is a good one to do at the board. If a student's answers are right but slow, the multiply-back check in the orange box is what to insist on next time, not more questions.

E4 Divide – these do not

- (a) $74 \div 8 = 9 \text{ r } 2$ (b) $100 \div 7 = 14 \text{ r } 2$ (c) $85 \div 9 = 9 \text{ r } 4$
 (d) $137 \div 5 = 27 \text{ r } 2$ (e) $191 \div 6 = 31 \text{ r } 5$ (f) $250 \div 8 = 31 \text{ r } 2$

Two errors to separate here. A remainder *bigger than the divisor* (writing $74 \div 8 = 8 \text{ r } 10$) means the quotient is one too small and they have stopped early – that is a method error. A remainder that is simply wrong by one or two is arithmetic. Only the first needs re-teaching.

Nobody should be writing a decimal here; if they do, it is worth saying that the remainder is the useful form in this unit, because the remainder is what decides whether something is a factor.

E5 Find the missing number

- (a) 9 (b) 12 (c) 9 (d) 56 (e) 14 (f) 13 (g) 12 (h) 100

The four with the gap on the left of a division – (d) and (h) – are the hard ones, because the missing number is the *big* one and you have to multiply to find it. Expect students to divide out of habit and get $8 \div 7$. The fix is the orange box: say the fact family out loud.

(f) $13 \times 9 = 117$ is outside the tables on purpose; it is there to be worked out rather than recalled.

E6 Use it: is it a factor?

$84 \div 6 = 14$ – yes

$92 \div 8 = 11 \text{ r } 4$ – no

$126 \div 9 = 14$ – yes (already done in E3d)

$100 \div 7 = 14 \text{ r } 2$ – no (already done in E4b)

The two repeats are deliberate. A student who does all four from scratch has not read the instruction; a student who spots them has done the thing this sheet is really for, which is noticing that the same division keeps turning up. Ask who found them before you give the answers.

Note that rows 3 and 4 both give 14 — one with nothing left over and one with a remainder of 2. That pair is the whole idea of a factor in two lines, and it is worth pointing at.

E7 Word problems

1. $6 \times 24 = 144$ markers.
2. $96 \div 8 = 12$ sheets each.
3. $8 \times 12 = 96$ slices altogether. $96 \div 29 = 3 \text{ r } 9$, so each student gets **3 slices** and Mr Bowen eats **9**.

Problem 3 is the deadpan one. Read it completely straight and do not explain the joke — the maths is correct, which is the point. It is also the only two-step question on the sheet: multiply first to find the total, then divide with a remainder. A student who divides 8 by 29, or 12 by 29, has missed the first step; send them back to the first sentence rather than to the arithmetic.